24CSA111 - DATABASE MANAGEMENT SYSTEM (DBMS)

DBMS – Unit 2

- Keys
- CODD's Rules
- Design Issues
- ER Model
- Attribute Type
- Weak Entity Sets
- Extended ER Features
- ER to Relational Mapping
- Structure of Relational Databases

Keys

- Key plays an important role in relational database; it is used for identifying unique rows from table. It also establishes relationship among tables.

- Types of keys
  - Primary key
  - Super key
  - Candidate key
  - Alternate key
  - Composite key
  - Foreign key

CODD's Rules

Rule 1: The Information Rule – All data should be presented in a table form.

Rule 2: Guaranteed Access Rule- All data should be accessible without ambiguity.

Rule 3: Systematic Treatment of Null values-A field should be allowed to remain empty.

Rule 4: Dynamic On-Line Catalog based on the Relational Model-Storing the structure definition in special system tables.

Rule 5: Comprehensive Data Sublanguage Rule-SQL

CODD's Rules

Rule 6: View updating rule- Data manipulation that has direct access to a table.

Rule 7: High level insert, update and delete-DML

Rule 8: Physical Data Independence

Rule 9: Logical Data Independence

Rule10:Integrity Independence – using constraints.

CODD's Rules

Rule 11: Distribution Independence- A user should be totally unaware of whether or not the database is distributed.

Rule 12: Non subversion Rule-There should be no way to modify the database structure other than through the multiple row database language (like SQL).

Design Issues

- Use of Entity sets vs Attributes (e.g. telephone attribute in employee entity)
- Use of Entity sets vs Relationship sets (e.g. there will be situations whether an object can be best expressed by an entity set or a relationship set e.g. bank loan)
- Binary vs nary relationship sets (between two entiites and more than two entities)

Data Model

A **data model**—a collection of concepts that can be used to describe the structure of a database, provides the necessary means to achieve this abstraction.

structure of a database, means the data types, relationships, and constraints that apply to the data.

ER Model

The ER model describes data as *entities*, *relationships*, and *attributes.*

The basic object that the ER model represents is an **entity**, which is a *thing* in the real world with an independent existence. An entity may be an object with a physical existence (for example, a particular person, car, house, or employee) or it may be an object with a conceptual existence (for instance, a company, a job, or a university course).

Each entity has **attributes**—the particular properties that describe it. For example, an EMPLOYEE entity may be described by the employee's name, age, address, salary, and job.

ER Relationship Diagram

An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system.

Relationship Types:

- one to one
- one to many
- many to many

Summary of notation for ER diagrams

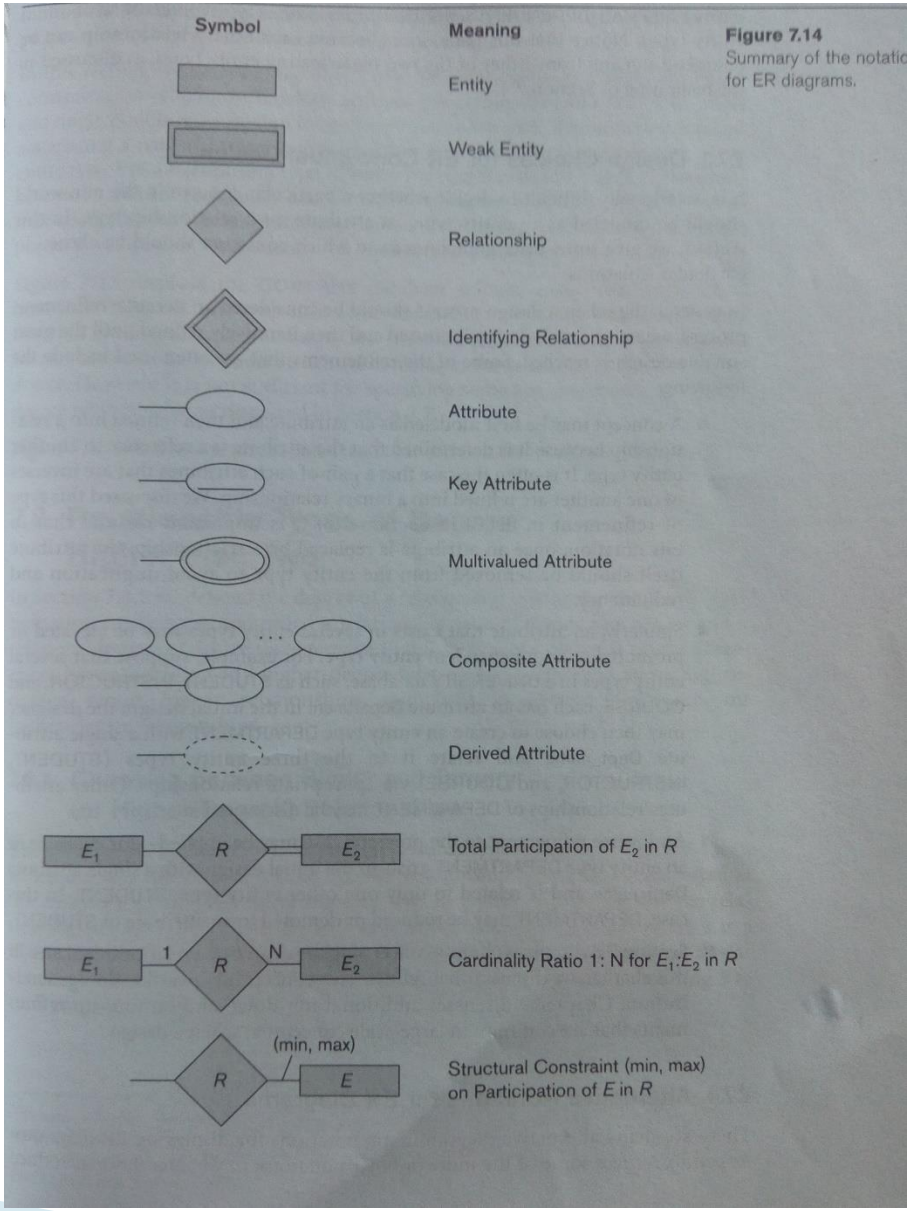
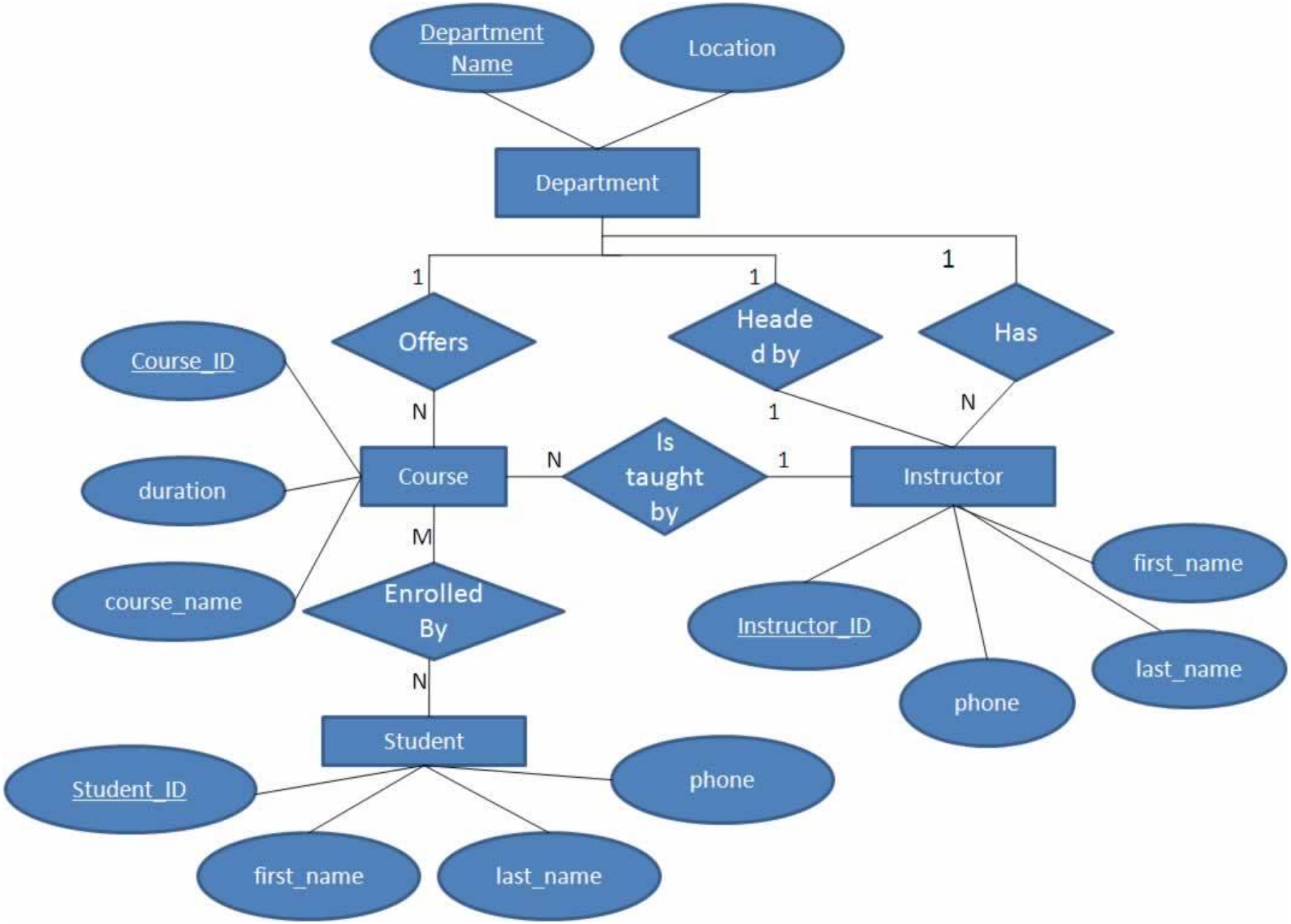


Figure 7.14
Summary of the notatio
for ER diagrams.

ER Model


Department Name
Location
Department
1
1
1
Offers
Heade d by
Has
Course_ID
N
1
N
Course
N
Is taught by
1
Instructor
duration
M
course_name
Enrolled By
first_name
Instructor_ID
last_name
phone
N
Student
Student_ID
phone
first_name
last_name

Attribute Types

The basic object that the ER model represents is an entity, which is a thing in the real world with an independent existence.

Each entity has attributes, the particular properties that describe it.

For example, an student entity may be described by the student name, course, address and contact number.

Attribute Types

- Simple
- Composite (e.g. Address)
- Single valued (e.g. age)
- Multivalued (e.g. college_degrees)
- Stored (e.g. DOB)
- Derived (e.g. age)

Weak Entity Sets

Entity types that do not have key attributes of their own are called **weak entity types**.

Entities belonging to a weak entity type are identified by being related to specific entities from another entity type in combination with one of their attribute values.

A weak entity type normally has a **partial key**, which is the attribute that can uniquely identify weak entities that are *related to the same owner entity.*

Extended ER Features

The Extended Entity-Relationship Model is a more complex and high-level model that extends an E-R diagram to include more types of abstraction, and to more clearly express constraints.

The EER model includes *all the modelling concepts of the ER model*. In addition, it includes the concepts of **subclass** and **superclass** and the related concepts of **specialization** and **generalization.**

Extended ER Features

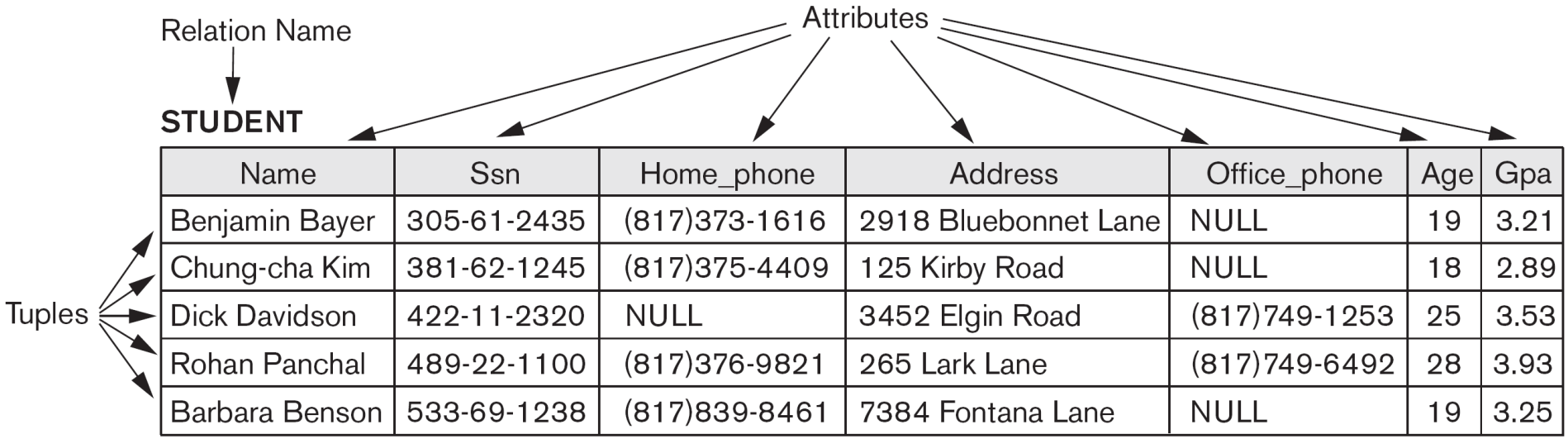
Another concept included in the EER model is that of a **category** or **union type,** which is used to represent a collection of objects (entities) that is the *union* of objects of different entity types. Associated with these concepts is the important mechanism of **attribute and relationship inheritance**.

ER to Relational Mapping

- Step 1: Mapping Regular entity types
- Step 2: Mapping of Weak entity types
- Step 3: Mapping of Binary 1:1 relation types
- Step 4: Mapping of Binary 1:N relation types
- Step 5: Mapping of Binary M:N relation types
- Step 6: Mapping of Multivalued attributes
- Step 7: Mapping of N-ary Relationship types

Structure of Relational Databases

- Attribute
- Tuple
- Relation



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Figure 3.1
The attributes and tuples of a relation STUDENT.

THANK YOU